

Assignment 3 - IoT Sensor Data Analysis (Q10 correction)

Q1. Load the dataset.

Answer: `df = pd.read_csv("iot_sensor_data_raw.csv")`

Q2. Display its dimensions and structure.

Answer: 20,000 rows x 9 columns: Timestamp, Device_ID, Temperature, Humidity, Pressure, Vibration, Battery_Level, Location, Machine_Status.

Q3. Convert Timestamp into Pandas datetime.

Answer: `df["Timestamp"] = pd.to_datetime(df["Timestamp"], errors="coerce")`. All 20,000 timestamps parse.

Q4. Check whether the timestamps are correctly ordered.

Answer: No. The file is not chronological; sort by Device_ID and Timestamp before time-series analysis.

Q5. Identify missing sensor readings.

Answer: Temperature 299; Humidity 300; Pressure 300; Vibration 298. Other fields have no missing values.

Q6. Calculate percentage of missing values for each sensor.

Answer: Temperature 1.50%; Humidity 1.50%; Pressure 1.50%; Vibration 1.49%; Battery_Level 0%.

Q7. Handle missing sensor values using an appropriate method.

Answer: Sort by device/time and impute each missing sensor with that device's median. This fills all missing sensor readings and limits cross-device bias.

Q8. Identify abnormal temperature readings.

Answer: Assumed warning >80 C and critical >100 C: 2,053 warning and 54 critical readings; 2,107 exceed 80 C.

Q9. Identify abnormal vibration readings.

Answer: Assumed warning >5 and critical >10 dataset units: 9 warning and 50 critical readings.

Q10. Identify machines with critically low battery levels.

Answer: Using the specified Battery_Level <20 cutoff, these low-battery readings occur on all eight devices: DEV_001 286; DEV_002 301; DEV_003 269; DEV_004 310; DEV_005 310; DEV_006 297; DEV_007 257; DEV_008 287. Total: 2,317 readings.

Q11. Explain the abnormal-reading thresholds.

Answer: Battery limits follow the assignment. Temperature >80 C / >100 C and vibration >5 / >10 are provisional Warning/Critical cutoffs, not manufacturer specifications; confirm them with equipment engineering before operational use.

Q12. Calculate average temperature by hour.

Answer:

Mean Celsius by hour 00-23: 70.08, 70.54, 69.93, 69.72, 69.81, 70.26, 70.25, 70.65, 70.39, 70.97, 69.93, 69.89, 69.99, 70.13, 69.84, 70.22, 69.97, 70.55, 70.22, 70.00, 70.14, 70.14, 71.17, 70.40.

Q13. Calculate average temperature for each device.

Answer:

DEV_001 70.25 C; DEV_002 70.38 C; DEV_003 69.93 C; DEV_004 70.58 C; DEV_005 70.11 C; DEV_006 70.00 C; DEV_007 70.14 C; DEV_008 70.32 C.

Q14. Calculate average vibration for each device.

Answer:

DEV_001 2.553; DEV_002 2.564; DEV_003 2.539; DEV_004 2.524; DEV_005 2.494; DEV_006 2.553; DEV_007 2.521; DEV_008 2.528.

Q15. Find the maximum temperature recorded by each device.

Answer:

DEV_001 159.75 C; DEV_002 151.82 C; DEV_003 157.60 C; DEV_004 151.97 C; DEV_005 152.97 C; DEV_006 136.00 C; DEV_007 153.88 C; DEV_008 153.62 C. These extreme maxima merit sensor validation.

Q16. Find the minimum battery level for each device.

Answer:

DEV_001 2.26%; DEV_002 2.15%; DEV_003 3.67%; DEV_004 8.51%; DEV_005 2.96%; DEV_006 2.20%; DEV_007 2.15%; DEV_008 2.01%.

Q17. Determine which device has the highest average vibration.

Answer:

DEV_002, approximately 2.564.

Q18. Determine which factory has the highest average temperature.

Answer:

By Location: Factory_A 70.23 C; Factory_B 70.15 C; Factory_C 70.27 C. **Factory_C** is highest.

Q19. Create Battery_Status: >=50 Healthy; 20-49 Moderate; <20 Critical.

Answer:

```
df["Battery_Status"] = np.select([df.Battery_Level < 20, df.Battery_Level < 50], ["Critical", "Moderate"], default="Healthy")
Counts: Healthy 10,978; Moderate 6,705; Critical 2,317.
```

Q20. Create a Temperature_Status column.

Answer:

Normal <=80 C; Warning >80 and <=100 C; Critical >100 C. Counts: Normal 17,893; Warning 2,053; Critical 54.

Q21. Create a Vibration_Status column.

Answer:

Normal <=5; Warning >5 and <=10; Critical >10. Counts: Normal 19,941; Warning 9; Critical 50.

Q22. Create overall Machine_Health (Normal, Warning, Critical) from suitable sensor conditions.

Answer:

Assign Critical if any sensor is Critical; otherwise Warning if temperature/vibration is Warning or battery is Moderate; otherwise Normal. Counts: Normal 9,824; Warning 7,771; Critical 2,405.

Q23. Find devices that experienced critical conditions more than five times.

Answer:

Using composite Critical = battery <20 OR temperature >100 C OR vibration >10, all eight qualify. Critical readings: DEV_001 304; DEV_002 312; DEV_003 276; DEV_004 323; DEV_005 319; DEV_006 305; DEV_007 270; DEV_008 296.

Q24. Find the factory with the highest number of abnormal sensor readings.

Answer:

Count records with overall Machine_Health Warning or Critical: Factory_A 3,440; Factory_B 3,371; Factory_C 3,365. **Factory_A** is highest.

Q25. Calculate percentage of time each device operates under Warning/Critical conditions.

Answer:

Using sampled records as a proxy for operating time: DEV_001 50.00%; DEV_002 51.86%; DEV_003 50.21%; DEV_004 51.40%; DEV_005 50.80%; DEV_006 51.93%; DEV_007 50.26%; DEV_008 50.58%. DEV_006 has the highest non-normal fraction.

Q26. Identify the device that requires the highest maintenance priority.

Answer:

Rank by number of composite critical records: **DEV_004** with 323 critical readings (highest). If prioritizing total warning/critical exposure instead, DEV_006 is highest at 51.93%; final maintenance triage should use severity and device specifications.